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Central Ideas and Details

10 questions · ~15 min

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INFORMATION AND IDEAS

In present-day Chiapas, Mexico, archaeologist Robert Rosenswig, remote-sensing specialist Ricardo López-Torrijos, and colleagues have located 41 smaller settlements surrounding the ancient Mesoamerican city of Izapa. The researchers have concluded that these settlements were culturally linked to Izapa because each of the settlements is the same age and configured in the same manner as Izapa, with a pyramid to the north and a plaza to the south. Their shared structural orientation suggests that residents of the settlements likely performed some of the same cultural ceremonies as residents in Izapa did.

Which choice best states the main idea of the text?

- A. Researchers have determined that the arrangement of Izapa's structures was based on those of other nearby settlements.
- B. Cultural ceremonies in Izapa seem to have played a more important role for its residents than those in smaller, surrounding settlements did.
- C. Although archaeologists have learned much about Izapa over years of research, they have only recently found the smaller settlements that surrounded it.
- D. Researchers have inferred that Izapa was related to the smaller settlements that surrounded it based in part on the similarity of their construction.

Paleontologist Alan Tennyson and colleagues studied fossil bones found in New Zealand that are more than 55 million years old. The researchers determined that the fossil bones belonged to two previously unknown species of prehistoric penguins. Moreover, they estimated that one of the two penguin species was more than three times the size of the emperor penguin, which is the largest penguin species that exists today.

Based on the text, which choice best describes the two previously unknown penguin species?

- A. They are frequently studied by paleontologists.
- B. They are no longer living species.
- C. They were smaller than penguin species that exist today.
- D. They spent little of their lives in water.

Paleontologists searching for signs of ancient life have found many fossilized specimens of prehistoric human ancestors, including several from the Pleistocene era discovered in a geological formation in the Minatogawa quarry in Japan. However, to study the emergence of the earliest multicellular organisms to appear on Earth, researchers must turn elsewhere, such as to the Ediacaran geological formation at Mistaken Point in Canada. A UNESCO World Heritage Site, the 146-hectare reserve contains more than 10,000 fossils that together document a critical moment in evolutionary history.

What does the text indicate about the geological formation at Mistaken Point?

- A. It holds a greater number of fossils but from a smaller variety of species than the formation in the Minatogawa quarry does.
- B. It has provided evidence that the earliest human species may have emerged before the Pleistocene era.
- C. It is widely considered by paleontologists to be the most valuable source of information about prehistoric life forms.
- D. It contains specimens from an older time period than those found in the formation in the Minatogawa quarry.

A contraction of “you all,” the pronoun “y’all” has long been used as a plural version of “you” in the South and in Black communities around the US. In recent decades, most other English-speaking communities in the US have begun to use “y’all.” What explains its rise in popularity? Many varieties of English have no pronoun that specifically addresses more than one person and instead must use “you” to address both one person and more than one. But “y’all” always refers to two or more people. As a result, it conveys the speaker’s meaning more precisely than “you” can.

Which question does the text most directly attempt to answer?

- A. How many other plural versions of the pronoun “you” are there in English, besides “y’all”?
- B. Why has the pronoun “y’all” become more widely used in the US?
- C. When was the first recorded use of the pronoun “y’all” in the English language?
- D. Is “y’all” commonly used in English-speaking regions of the world besides the US?

NASA's Aspera mission, led by Carlos Vargas, will investigate the circumgalactic medium (CGM), the huge swaths of low-density gas that fill and surround galaxies. Specifically, the team will focus on portions of the gas that exist in a "warm-hot" phase: these portions haven't previously been observable but are thought to fuel new star formation and hold most of the mass that makes up a galaxy. Using a telescope capable of revealing these parts of the CGM, the Aspera mission should help answer long-standing questions about how galaxies emerge, change, and even interact.

Which choice best states the main idea of the text?

- A. As the leader of NASA's Aspera mission, Vargas will be the first person to investigate the makeup of the CGM.
- B. Although galaxies that are surrounded by the CGM have been studied, researchers have been unable to directly observe low-density gas in the CGM in the "warm-hot" phase.
- C. Researchers don't yet have a complete understanding of the process of galaxy evolution but have raised the possibility that galaxies interact with each other at times.
- D. The Aspera mission is expected to produce the first direct observations of CGM gas in the "warm-hot" phase, which likely has an important role in the evolution of galaxies.

In a study by Mika R. Moran, Daniel A. Rodriguez, and colleagues, residents of Quito, Ecuador, and Lima, Peru, were surveyed about parks in their cities. Of the 618 respondents from Quito, 82.9% indicated that they use the city's parks, and of the 663 respondents from Lima, 72.7% indicated using city parks. Given that the percentage of Quito respondents who reported living within a 10-minute walk of a park was much lower than that reported by Lima respondents, greater proximity alone can't explain the difference in park use.

The text makes which point about the difference between the proportions of Quito residents and Lima residents using parks?

- A. It was much larger than the researchers conducting the study expected.
- B. It is caused by something other than the parks' proximity to city residents.
- C. It could be due to inaccuracies in the survey results.
- D. It was calculated using sources that predate the survey.

At over a thousand pages across two volumes, *The Fifty-Year Mission*, compiled by Edward Gross and Mark A. Altman, is presented as the “complete, uncensored, unauthorized oral history” as told by the people behind the media franchise *Star Trek*. The work aspires to be comprehensive by, for example, including accounts from cast and crew members of every *Star Trek* television series and film to date. But while *The Fifty-Year Mission* is clearly a unique and valuable resource, it has a shortcoming common among oral histories: it lacks a clear authorial point of view that could otherwise unite the various accounts into a cohesive whole.

Which choice best states the main idea of the text?

- A. The compilers of *The Fifty-Year Mission* had lofty goals for their oral history of the *Star Trek* franchise, but the published work lacks information about many key events in the franchise’s history.
- B. *The Fifty-Year Mission* includes more accounts from people involved with *Star Trek* television shows than it does from people involved with *Star Trek* films.
- C. The large amount of material compiled into *The Fifty-Year Mission* is surprising given that many of the people involved in the *Star Trek* franchise did not participate in the oral history project.
- D. *The Fifty-Year Mission* represents a worthwhile attempt to thoroughly recount the history of the *Star Trek* franchise, but its approach has an important limitation.

Using the Stratospheric Observatory for Infrared Astronomy (SOFIA), a team of astronomers mapped out the magnetic field of G47, one of the Milky Way's galactic bones (dense clouds of gas and dust that run through the middle of the arm of a spiral galaxy). Surprisingly, the map revealed a magnetic field with no clear pattern or direction. The researchers had expected the magnetic field to be similar to the more uniform fields seen in galactic bones in other arms of the Milky Way.

According to the text, what was surprising about the researchers' mapping of the magnetic field of galactic bone G47?

- A. It showed a weaker magnetic field than expected.
- B. It implied that previous mappings of the magnetic field were inaccurate.
- C. It produced magnetic field measurements similar to those for other galactic bones.
- D. It revealed a magnetic field that wasn't uniform.

In many of his sculptures, artist Richard Hunt uses broad forms rather than extreme accuracy to hint at specific people or ideas. In his first major work, *Arachne* (1956), Hunt constructed the mythical character Arachne, a weaver who was changed into a spider, by welding bits of steel together into something that, although vaguely human, is strange and machine-like. And his large bronze sculpture *The Light of Truth* (2021) commemorates activist and journalist Ida B. Wells using mainly flowing, curved pieces of metal that create stylized flame.

Which choice best states the text's main idea about Hunt?

- A. He often depicts the subjects of his sculptures using an unrealistic style.
- B. He uses different kinds of materials depending on what kind of sculpture he plans to create.
- C. He tends to base his art on important historical figures rather than on fictional characters.
- D. He has altered his approach to sculpture over time, and his works have become increasingly abstract.

To protect themselves when being attacked, hagfish—jawless marine animals that resemble eels—will release large quantities of slimy, mucus-like threads. Because these threads are unusually strong and elastic, scientist Atsuko Negishi and her colleagues have been trying to recreate them in a lab as an eco-friendly alternative to petroleum-based fibers that are often used in fabrics. The researchers want to reproduce the threads in the lab because farming hagfish for their slime would be expensive and potentially harmful to the hagfish.

Which choice best states the text’s main idea?

- A. The slimy threads that hagfish release might help researchers create a new kind of fabric.
- B. Hagfish have inspired researchers to develop a new petroleum-based fabric.
- C. Hagfish are not well suited to being raised in captivity.
- D. The ability of hagfish to slime their attackers compensates for their being jawless.